LECTURE 3
BATTERY MANAGEMENT SYSTEM

Batteries can be configured in series, in parallel, or in a combination of both. In electric and hybrid vehicles, the battery pack is the energy-storage device and consists of individual electrochemical cells connected in a series string to deliver the required voltage. Strings of series-connected cells can then be connected in parallel to increase the capacity of the storage system. A battery pack also includes electronic circuits, typically located outside the battery pack in the arrangement described in the source notes. These circuits control charging, discharging and the balanced utilization of battery cells. The electronic circuit, together with its controller hardware and software algorithms, manages the battery pack and protects the cells. The primary purpose of battery-pack management is to prevent cells from operating outside their safe region, helping extend battery life and minimize replacement costs.

Battery-pack management techniques are applicable to energy-storage systems such as electric and hybrid vehicles, distributed power-generation units and portable consumer electronics. Hybrid vehicles place particularly demanding requirements on energy storage because their batteries undergo pulsed charge and discharge cycles. Consequently, battery management in hybrid vehicles needs to be advanced. The battery-management methods and cell-balancing methods described in these notes are also applicable to an ultracapacitor bank used as an energy-storage device; ultracapacitor cells can be connected in series to provide the desired voltage

BATTERY MANAGEMENT SYSTEM

A Battery Management System (BMS) is an electronic technology used to monitor, manage, and control the operation of a battery system. It consists of various sensors, actuators, controllers, signal lines, and embedded algorithms that work together to ensure the safe, reliable, and efficient operation of the battery. In electric vehicles (EVs), the BMS plays an important role because the battery is the primary source of electrical energy for vehicle propulsion.

The main objective of a BMS is to ensure battery safety and reliability, provide battery-state information required by the vehicle control unit (VCU) and energy management system, and take appropriate corrective actions when abnormal operating conditions occur. The BMS continuously collects real-time information about the battery pack and individual cells, including terminal voltage, current, and temperature, through suitable sensing and sampling circuits. It processes this information using embedded algorithms to estimate the condition of the battery and determine whether it is operating within safe limits.

The BMS estimates important battery parameters such as the State of Charge (SOC), State of Health (SOH), State of Power (SOP), and Remaining Useful Life (RUL). These estimates are communicated to the vehicle control unit to support energy management, power distribution, and charge–discharge control. Therefore, the BMS helps improve battery safety, performance, reliability, and service life. The BMS protects the battery against overcharge, overdischarge, short circuits and thermal abuse.

The measurements, parameter estimations and outputs generated in a BMS are shown in Figure1.

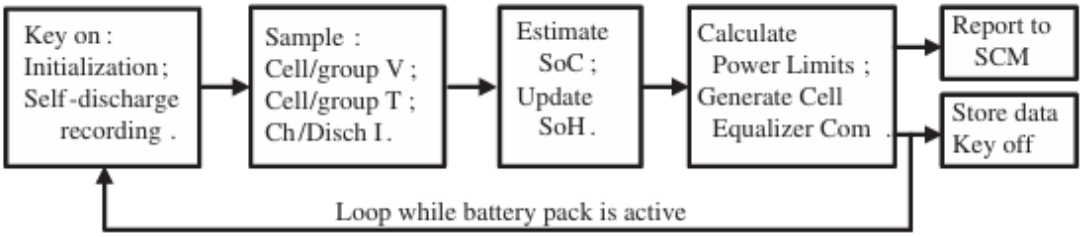


Figure 1: Parameter estimation and pack management in a BMS

The BMS initializes when the system is powered on, such as when the key is turned on in a hybrid electric vehicle. During initialization, it records self-discharge during the period when the system was off. If self-discharge is excessive, this information is reported to the SOH-monitoring algorithm. Other battery-management parameters are estimated during each measurement cycle while the pack is operating.

SOC indicates the available capacity of a battery. It is important for battery-pack protection and for vehicle powertrain control. SOC should be maintained within a specified band to enhance battery life. In sophisticated management systems, the SOC of individual cells or groups of cells is determined to check whether SOC is distributed uniformly across the pack. SOC is typically expressed as a percentage of rated capacity rather than maximum available capacity, which may be lower because of aging and environmental effects. It may also be calculated using maximum available capacity. This approach can be useful for cell equalization because cells in a series string generally experience the same environment.

SOH describes the working condition of the battery pack and measures its ability to deliver power compared with a new pack. A cell's capacity and power may fade with aging, and differences between cells can indicate deterioration. Cell capacity and other parameters are used by an algorithm to estimate pack SOH. SOH information supports battery safety and helps determine how much power the pack can deliver within its capability. Estimation algorithms compare measured and estimated cell parameters with reference values or with neighbouring cells. Abnormal cell voltage or SOC compared with nearby cells may indicate poor cell health. Excessive pack self-discharge is also checked against preset limits. SOH information can help identify and replace damaged cells instead of replacing the entire pack.

Temperature is a primary environmental factor affecting the SOC of an energy-storage system. Temperature differences among cells can create SOC imbalances and can also change self-discharge rates. Although pack thermal management is part of the cooling-system design, cell-temperature information must be used effectively for cell protection and health monitoring.The BMS calculates the maximum power available from the battery at a given time using SOC and terminal-voltage information while ensuring that voltage, current, SOC and other design limits are not violated. It sets charge and discharge power limits to protect the battery and reports these limits to the supervisory controller for powertrain control in electric and hybrid vehicles.

Battery-management systems can operate at three levels: pack-level management, modular-pack-level management and cell-level management. Pack-level management is the most basic arrangement and monitors overall pack voltage and SOC. It does not provide the same degree of protection to individual cells as monitoring their individual parameters. In modular-pack-level management, groups of cells are treated as modules, and the BMS uses group voltage, current and temperature measurements. Cell-level management monitors individual-cell parameters and permits the most complete cell equalization and balancing. When cell-equalizer

circuits are installed, the BMS generates command signals based on its measurements and estimations, and the circuits act on those signals to balance cells or groups of cells.

Data logging is another important function of the energy-storage management system. Voltage, current, temperature, SOC and the number of charge-discharge cycles can be stored as functions of time. These records support SOH monitoring, diagnostics and fault analysis.

MAIN FUNCTIONS OF A BATTERY MANAGEMENT SYSTEM

The main functions of a BMS include data collection, state monitoring, safety protection, charging control, energy management, equalization management, thermal management, and information management. These functions operate together to maintain the battery within its permitted operating conditions and to provide the information required for effective battery operation.

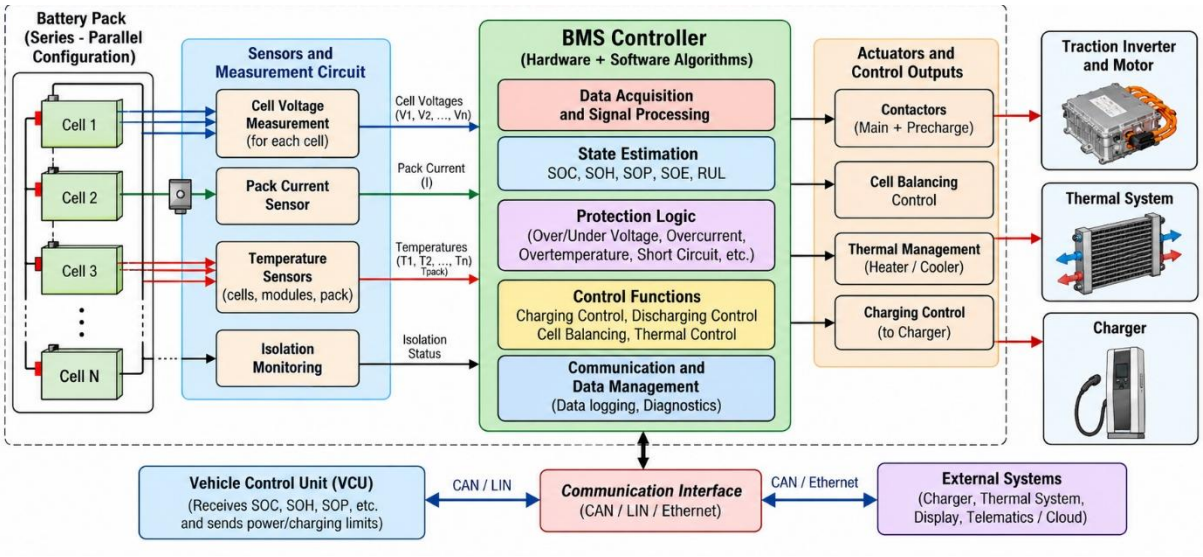


Figure 2: Battery Management System

1. Data Collection
   Data collection is the first and most fundamental function of a BMS. To determine the operating condition of the battery accurately, the BMS continuously collects information from the battery pack and its individual cells through sensing and sampling circuits. The main parameters measured are cell voltage, pack voltage, battery current, and temperature. Depending on the battery system, additional information may also be collected from sensors and other electronic components.
   The collected data is supplied to the BMS controller for processing and analysis. Accurate, real-time measurements are essential because the battery's operating condition changes continuously during charging, discharging, acceleration, regenerative braking, and variations in ambient temperature. The quality of the collected data directly affects the accuracy of battery-state estimation, protection functions, and energy management decisions.
2. State Monitoring
   A battery is a complex, nonlinear, time-varying system whose internal condition changes during operation. Consequently, its condition cannot be determined only by measuring terminal voltage and current. The BMS uses embedded algorithms and

estimation techniques to determine the battery's important operating states from real-time measurements.

The principal parameters estimated by the BMS are SOC, SOH, SOP, and State of Energy (SOE). SOC indicates the remaining charge in the battery relative to its usable capacity. SOH represents the battery's condition compared with its original or reference condition, particularly in terms of capacity degradation and internal resistance changes. SOP indicates the power that the battery can safely deliver or accept under specified operating conditions. SOE represents the remaining usable energy available in the battery.

These estimates help the vehicle control unit understand the battery's current condition and make suitable decisions regarding vehicle operation, charging, discharging, and power distribution. Accurate state monitoring also supports the prevention of excessive charging or discharging and helps improve battery utilization.

3. Safety Protection

Safety protection is one of the most important functions of a BMS. It involves continuous fault detection, diagnosis, and the implementation of suitable protective actions to prevent damage to the battery and associated electrical components. Since battery cells operate within specified voltage, current, and temperature limits, the BMS monitors these parameters continuously and identifies abnormal operating conditions.

Common battery faults include overvoltage, undervoltage, overcurrent, excessively high temperature, short circuits, loose electrical connections, reduced insulation, and electrolyte leakage. Overvoltage may occur during excessive charging, whereas undervoltage may result from excessive discharging. Overcurrent and short circuits can generate excessive heat and damage the battery or connected components. Reduced insulation can create electrical safety risks, particularly in high-voltage EV battery packs. Faults may also occur in electronic components such as sensors, actuators, and controllers.

When an abnormal condition is detected, the BMS initiates an appropriate protective response according to the system design. This may include limiting the charging or discharging current, issuing a warning, stopping the charging process, or disconnecting the battery from the vehicle's electrical system through contactors. Thus, safety protection helps prevent hazardous operating conditions and improves the reliability of the complete battery system.

4. Charging Control

Charging control is essential because the charging process directly affects battery safety, performance, and service life. The BMS controls or coordinates battery charging according to the battery's real-time operating condition, temperature, voltage limits, and the power available from the charger. Its charging-management function ensures that the battery receives an appropriate amount of current and voltage under the prevailing conditions.

During charging, the BMS monitors individual cell voltages, pack voltage, battery current, and temperature. If a cell approaches its maximum permitted voltage or the battery temperature exceeds its specified limit, the BMS can request a reduction in charging current or stop charging, depending on the severity of the condition. It may also communicate charging limits to the charger or vehicle control unit.

The BMS therefore coordinates the charging process to reduce the risk of overcharging and overheating. It can also support cell balancing during charging so that differences

in cell voltage and charge level are reduced. The exact charging algorithm depends on the battery chemistry, charger design, and battery manufacturer's operating limits.

5. Energy Management
   Energy management involves controlling the energy delivered by the battery and coordinating energy recovery during regenerative braking. In an electric vehicle, the BMS uses the measured battery parameters and estimated battery states to determine suitable operating limits for charging and discharging. These limits help ensure that the battery operates safely while meeting the energy requirements of the vehicle.
   During acceleration or other high-power operating conditions, the battery must supply the required electrical power without exceeding its permitted current, voltage, or temperature limits. During regenerative braking, the electric motor operates as a generator and converts part of the vehicle's kinetic energy into electrical energy. The BMS determines whether the battery can safely accept this recovered energy and communicates the applicable charging or power limits to the vehicle control system.
   Energy management also involves optimizing energy distribution according to the condition of the energy-storage system. By coordinating battery power delivery and energy recovery, the BMS supports efficient vehicle operation, protects the battery from excessive loading, and helps improve the overall utilization of stored energy.
6. Equalization Management (Cell Balancing)
   Battery packs consist of multiple cells connected in series and parallel to obtain the required voltage and capacity. However, individual cells may not have exactly identical characteristics because of variations during manufacturing, transportation, storage, ageing, and operation. Differences in cell capacity, internal resistance, and self-discharge can cause the cells to develop unequal voltage and charge levels.
   Equalization management, commonly called cell balancing, is the process of reducing these differences between cells. Without adequate balancing, one cell in a series-connected string may reach its maximum permitted voltage before the other cells become fully charged. Similarly, a weaker cell may reach its minimum permitted voltage before the remaining cells are discharged. This limits the usable capacity of the complete battery pack and may create safety or reliability problems.
   The BMS monitors individual cell voltages and, where supported by the design, their estimated states of charge. It then applies an appropriate balancing method to reduce cell-to-cell differences. The two main methods are passive balancing and active balancing. Passive balancing dissipates excess energy from higher-charge cells, generally through resistors. Active balancing transfers energy from higher-charge cells to lower-charge cells or between cells and other storage elements. Cell balancing improves the usable capacity of the battery pack and helps maintain more consistent cell operation.
7. **Thermal Management**
   Battery performance and service life are strongly influenced by temperature. The battery is affected by ambient temperature as well as the heat generated internally during charging and discharging. If the battery operates outside its recommended temperature range, its performance, charging capability, ageing rate, and safety may be adversely affected.
   The BMS monitors temperature at suitable locations within the battery pack and uses this information to coordinate the thermal management system. Depending on the battery design and operating conditions, thermal management may involve active cooling, heating, or other temperature-control methods. The intensity of heating or

cooling is adjusted according to the measured temperature distribution, battery operating state, and charge–discharge requirements.

Maintaining an appropriate temperature range helps improve battery performance, reduce excessive thermal stress, and extend service life. In EV battery packs, effective thermal management is particularly important because cells may experience different temperatures due to their positions within the pack, differences in cooling conditions, and variations in electrical loading.

8. **Information Management**

Information management involves organizing, processing, storing, and communicating battery-related data. A BMS contains several functional modules that must coordinate their activities and exchange information reliably. The system therefore manages measurement data, battery-state estimates, fault information, protection commands, and communication between the battery controller and other vehicle systems.

The BMS filters and processes collected data to improve the usefulness of measurements and identify important operating conditions. It stores relevant information, such as fault records and battery operating history, where the system supports data logging. It also communicates with external electronic units, including the vehicle control unit and charging system, to exchange battery-state information and operating limits.

Advanced battery management systems may also communicate with cloud platforms for remote monitoring, data analysis, and battery-performance assessment. Such communication can support the identification of operating trends and potential battery-management problems. However, cloud connectivity is an additional capability and is not necessary for every BMS.

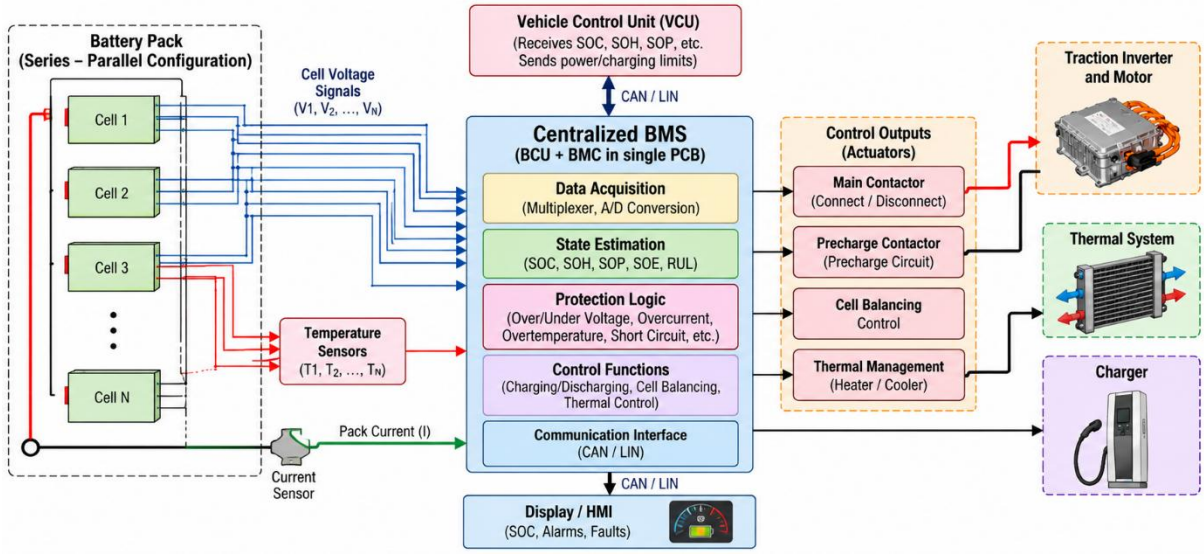
BMS TOPOLOGIES

The topology of a BMS refers to the arrangement and interconnection of its electronic monitoring and control components. The choice of topology directly affects system cost, reliability, installation convenience, maintenance, measurement accuracy, and the ability to expand the battery system.

The hardware of a BMS commonly includes a Battery Monitoring Circuit (BMC) and a Battery Control Unit (BCU). The BMC is primarily responsible for acquiring battery-cell measurements, such as voltage and temperature, and supporting cell-level monitoring. The BCU performs higher-level functions, including battery-state estimation, fault diagnosis, charge–discharge control, and communication with the vehicle control unit. Based on the structural relationship between the monitoring circuits, control unit, and battery cells, BMS topologies can be classified into two main types: centralized BMS and distributed BMS.

Centralized BMS Topology

A centralized BMS uses a single central electronic circuit board to perform battery monitoring and management functions. The measurement signals from the individual cells are transmitted to this common board, where the signals are processed and used for battery-state estimation, protection, and control. The central controller communicates with other systems, such as the vehicle control unit, display interface, and charging equipment, as required by the application.



This topology is generally suitable for battery systems with relatively low capacity, lower total voltage, and a comparatively small number of cells. Since the monitoring and control functions are integrated into one unit, the system can have a compact structure and relatively low cost. Communication within the central board can be fast, and the integrated design can reduce the number of separate electronic modules.

However, a centralized BMS requires a large number of wires and connectors to connect the individual cells to the central circuit board. As the battery pack becomes larger, the wiring becomes more complex, increasing installation and maintenance difficulties. The design must also provide sufficient electrical insulation and safety clearance between high-voltage and low-voltage circuits. This requirement can increase the size and complexity of the printed circuit board (PCB).

Another limitation is scalability. Since most monitoring functions are concentrated on one board, increasing the number of cells may require substantial changes to the wiring and circuit design. The concentration of functions in one unit can also create a single point of failure: if the central controller fails, several or all battery-monitoring functions may be affected.

Advantages of Centralized BMS

The main advantage of a centralized BMS is its relatively simple and compact architecture. Since the monitoring and control functions are integrated into a single unit, the number of separate electronic modules is reduced, which can lower the overall system cost. Communication between functions on the same board can be fast, supporting synchronized data collection. The smaller number of distributed electronic units can also simplify certain aspects of installation.

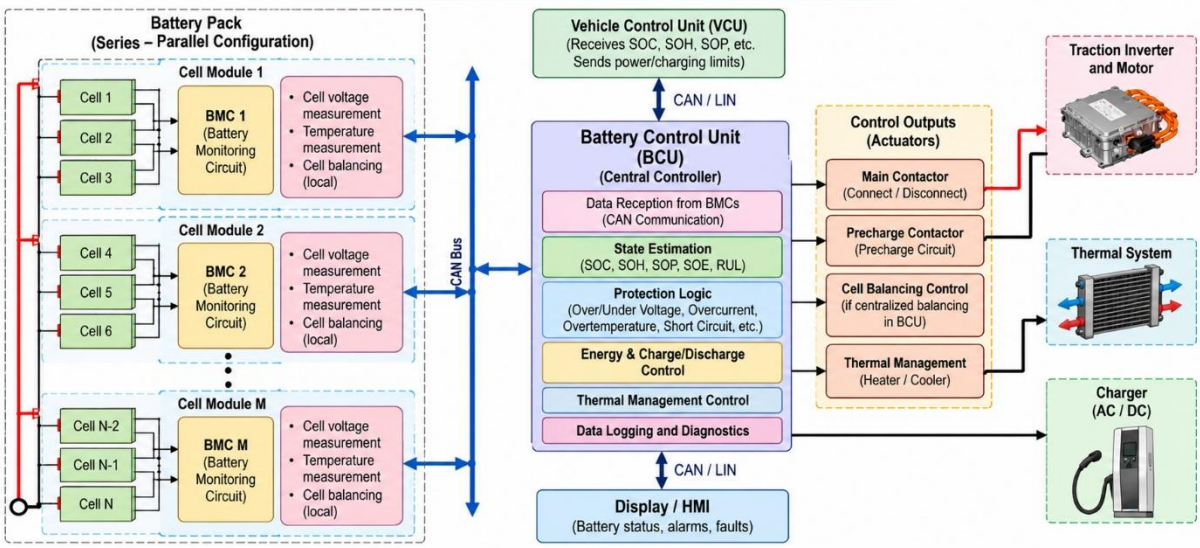
Disadvantages of Centralized BMS

The main disadvantage is the large number of wires and connectors required to connect

individual cells to the central board. This can make the wiring harness complicated and increase installation and maintenance effort. High-voltage safety requirements may require considerable spacing between measurement channels, increasing the PCB size. Furthermore, the integration of all functions into one board limits scalability and can make expansion to larger battery packs difficult. A failure in the central unit may also affect the operation of the entire BMS.

Distributed BMS Topology

A distributed BMS separates cell-level monitoring from higher-level battery control. In this arrangement, multiple Battery Monitoring Circuits (BMCs) are installed at different locations in the battery pack, while a central Battery Control Unit (BCU) coordinates the overall operation of the system.



Each BMC is connected to its corresponding cell or group of cells and monitors parameters such as cell voltage and temperature. Depending on the design, it may also support cell balancing and other local diagnostic functions. The BCU receives the measurement information from the BMCs, estimates battery states, detects faults, manages switching devices, coordinates charging and discharging, and communicates with the vehicle control unit.

The BMCs and BCU are connected through a communication network, commonly a Controller Area Network (CAN) bus. Each monitoring circuit acts as a communication node and sends its measurements and status information to the BCU. The network also enables the controller to communicate commands and operating limits to the monitoring units. Depending on the architecture, communication between monitoring units may also be supported.

Because the monitoring circuits are distributed throughout the battery pack, the wiring between each cell and its local monitoring circuit can be shorter than in a centralized design. This arrangement is particularly useful for large battery packs containing many cells or modules. It also allows the system to be expanded by adding monitoring units, provided that the communication network and controller have sufficient capacity.

A distributed BMS has a more modular structure than a centralized BMS. However, it requires additional electronic monitoring units and a reliable communication network. This can increase system complexity, component cost, and the effort required to diagnose communication faults.

Functions of the BMC and BCU

The Battery Monitoring Circuit (BMC) is responsible for collecting cell-level information, particularly voltage and temperature measurements. It may also perform local diagnostic and cell-balancing functions. Its main purpose is to provide accurate information about the condition of the cells it monitors.

The Battery Control Unit (BCU) performs pack-level management. It processes information received from the BMCs, estimates SOC and other battery states, identifies abnormal operating conditions, manages contactors or other switching devices, and coordinates charging and discharging. It also communicates with the vehicle control unit and other relevant systems.

Comparison of Centralized and Distributed BMS

Feature	**Centralized BMS**	**Distributed BMS**
Architecture	One central monitoring and control board	Multiple monitoring circuits with a central control unit
Cell monitoring	Measurement signals from cells go to one central board	Local monitoring circuits measure individual cells or cell groups
Wiring	More cell-level wiring to the central board	Shorter local cell connections, with inter-unit communication wiring
Cost	Generally lower for small battery packs	Generally higher due to additional monitoring units
Scalability	More difficult to expand	Generally easier to expand through additional monitoring units
Maintenance	Central-board faults may affect the complete system	Modular design can help isolate faults, depending on the architecture
Typical application	Relatively small battery systems	Larger battery packs with many cells or modules